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**Περιοδοντίτιδα και Πνευμονία Κοινότητας
Συστηματική ανασκόπηση**

Διπλωματική εργασία για την εκπλήρωση των απαιτήσεων

του προγράμματος μεταπτυχιακών σπουδών

Μεθοδολογία της Έρευνας στην Ιατρική και τις Επιστήμες Υγείας

Τζέμη Μαρία

Θεσσαλονίκη, 2024

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της Τζέμη Μαρίας

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Periodontitis and Community Pneumonia
Systematic review

MSc Thesis to fulfill the requirements
of postgraduate studies
Research Methodology in Medicine and Health Sciences

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ABBREVIATIONS

BOP – Bleeding on Probing

CAL – Clinical Attachment Level

CAP – Community-acquired Pneumonia

CP – Chronic Periodontitis

CHX- Chlorhexidine

PROLOGUE

1. Reasons for selection of the topic

Following a comprehensive analysis of the literature, we conclude that the infections of the respiratory system, such as pneumonia, and periodontitis may have a positive association. The same conclusion is observed in earlier systematic reviews that confirm the connection between oral hygiene and pneumonia episodes. Community-acquired pneumonia has not been covered in any of the systematic reviews that have been published so far on the link between periodontitis and different types of pneumonia. All the above elements prompted us to a new research hypothesis, a systematic review regarding periodontitis and community-acquired pneumonia, to comprehend all aspects of pneumonia. Positive correlations between periodontitis and community-acquired pneumonia may support an extensive strategy of social and preventative dental care for society.

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ABSTRACT

Introduction

Objective: The review presented aims to assess studies examining the relationship between periodontitis and community-acquired pneumonia (CAP) in adults, considering periodontitis as a possible contributory factor to the development of CAP.

Materials and Methods: The first search was conducted in 18 databases up to July 2022 and an update search up to April 2024 to identify studies answering the PECOS. The search was carried out for cross-sectional, cohort and case-control studies published within the last 10 years. Data extraction and quality assessment were done upon performing a narrative synthesis. The systematic review was registered in PROSPERO: CRD42024539085.

Results: Of the included studies from Japan, Brazil, and South Korea, a total of 364,378 subjects were included. Awano et al. (2008) reported that subjects with periodontal pockets (>4mm) faced increased mortality due to pneumonia, while De Melo Neto et al. (2013) pointed out that severe periodontitis was significantly associated with CAP; however, Kim et al. (2019), on the contrary stated that the possibility of such malignancies being conferred by diagnostic constraints and confounding adjustments could not be ruled out.

Discussion:

The studies examined have differences in study designs, population characteristics, and have differing diagnostic methods. Differing confounders like smoking and comorbidities may affect results, making them a subject of further exploration in understanding the mechanism linking periodontitis to CAP.

Conclusion:

More studies, having the same methodological criteria, are needed to clarify the relationship of periodontitis with CAP.

INTRODUCTION

1.1. Periodontitis

Periodontitis develops as an inflammatory disease, affecting the supporting structures of the teeth: the gingiva, periodontal ligament, cementum, and alveolar bone. (1) The onset is triggered by the interplay between the microbial biofilms popularly known as dental plaque and the adverse effect of the host's immune response, which leads to the progressive destruction of the periodontal tissues. (2,3) Usually, periodontitis starts in the form of gingivitis, a disorder where inflammation of the gingiva is reversible. However, if it remains untreated, periodontitis progresses to an irreversible state of attachment loss, destruction of alveolar bone, and, finally, tooth mobility and loss. (4)

Causes and Pathogenesis of Periodontitis

Causes of periodontitis are variable; the primary driving force may still be a bacterial infection, whereas some others like genetic predisposition, environmental influences like smoking, and systemic conditions like diabetes mellitus very significantly aid the progression of the disease. (5) Starting from a microbial biofilm consisting of gram-negative anaerobic bacteria including *Porphyromonas gingivalis*, (6) *Tannerella forsythia*, and *Treponema denticola*, the inflammatory process is set in motion. Tissue destruction and bone resorption result from host immune responses to these pathogens, especially the dysregulated production of pro-inflammatory cytokines. (7)

Immune Response in Periodontitis

Pathophysiologically periodontitis involves a complex immune response. Initially, neutrophils are recruited to the site of bacterial invasion to contain the infection. However, as the disease progresses, this response becomes chronic, with macrophages, T-cells, and B-cells releasing inflammatory mediators like interleukin-1 β (IL-1 β), tumor necrosis factor-alpha (TNF- α), and prostaglandins. (8) These mediators stimulate osteoclast activity, which leads to alveolar bone resorption and the progressive destruction of periodontal tissues.

Prevalence of periodontitis and Socioeconomic Impact

As a consequence of its significant prevalence and consequent impact on quality of life, periodontitis has emerged as a global health challenge. The numbers provided by the Global Burden of Disease Study, (2019) suggested that severe periodontitis is believed to affect almost one billion people worldwide, (9) thus classifying it as the sixth most common disorder globally. This situation increases the prevalence

with age and the ultimately heaviest burden of the disease over the age of 35 years. (10) Beyond being prevalent, periodontitis contributes considerably to the socio-economic burden. Loss of teeth due to this disease becomes an important hindrance to mastication, speech, and aesthetics and therefore impacts quality of life negatively. (11) The cost of managing periodontitis includes dentally related medication, treatment, and rehabilitation, which is an enormous financial burden on patients and healthcare systems alike. (12)

Periodontitis and Systemic Health

Interaction between oral health and systemic health has been the focus of increasing interest over the last few decades. Periodontal diseases, being chronic inflammatory diseases, have been linked to several systemic diseases, i.e., cardiovascular diseases, diabetes mellitus, adverse pregnancy outcomes, and respiratory infections. (13-15) The main underlying mechanism for causing these associations is the systemic dissemination of periodontal pathogens and their by-products. For instance, bacteria like *P. gingivalis* can enter systemic circulation and cause transient bacteremia. (16) Pro-inflammatory cytokines produced locally in periodontal tissues can also escape into the systemic circulation and cause chronic low-grade ascitic inflammation. (17)

The relationship between periodontitis and systemic health is well established. The presence of periodontal pathogens within atherosclerotic plaques, for instance, suggests a direct role in the pathogenic process of cardiovascular diseases. (18) Periodontitis has been shown to exacerbate systemic inflammation in diabetes mellitus, thus worsening glycemic control and creating a reciprocal dynamic between both conditions. (19) Of course, respiratory diseases such as pneumonia can affect periodontal health since oral bacteria get aspirated into the lower respiratory tract and thus may contribute to such infection. (20)

1.2. Community-acquired pneumonia (CAP)

Community-acquired pneumonia (CAP) is probably an everyday infection that may become life-threatening in a person who has not been hospitalized lately or in the setting of healthcare. (21) It involves acute infections and inflammatory responses in the lung parenchyma interfering with gas exchange and leading to respiratory symptoms. The CAP presentation can range from mild to severe depending on several factors (22) including the patient's age, comorbidities, and the organism responsible for the infection. (23) Common symptoms include fever, cough with or without sputum, pleuritic chest pain,

dyspnea, and fatigue. Clinically severe cases may, in turn, give rise to complications such as sepsis, respiratory failure, and multi-organ dysfunction. (24)

Pathogens Causing CAP

A variety of pathogens cause community-acquired pneumonia: bacterial, viral, and fungal. *Streptococcus pneumoniae* is the most commonly isolated bacterial pathogen; however, atypical pathogens such as *Mycoplasma pneumoniae*, *Legionella pneumophila*, and *Chlamydia pneumoniae* are also important. (25) Viral causes such as influenza and latterly SARS-CoV-2 have gained prominence. Diagnosis of CAP is commonly made through clinical judgment, with consideration of radiological findings such as infiltrates noted on chest X-ray. (26)

Global Burden of CAP

CAP is a major worldwide public health issue associated with noteworthy morbidity, mortality, and economic costs for its management. (27) Annually, CAP occurs in 1.5 to 10 cases per 1,000 adults in the general population, with a high incidence in children under five years and adults over 65 years. (28) Admission to the hospital for CAP is common, especially in developed countries, where it ranks as one of the dominant causes of admission. The burden is even heavier in low- and middle-income countries. It is a leading cause of mortality among children aged below five years. According to the World Health Organization, pneumonia contributed to about 15% of all deaths in children below five years of age in 2019, with most cases occurring in South Asia and Sub-Saharan Africa. (29)

Economic Impact of CAP

Apart from the obvious health effects imposed by CAP, there lie severe economic implications. The costs involved in the management of CAP would include, among others, hospital admissions, diagnostic workup (including chest X-ray), antimicrobial therapy, and loss of productivity. (30) The growing prevalence of antibiotic resistance rendered treatment more difficult, thus aggravating the financial burden on healthcare providers. CAP morbidity and mortality reputes to be very significant, especially in a vulnerable sector such as the elderly and those with comorbid conditions and impaired immunity. (31) The death rates among hospitalized patients suffering from CAP could still be approximated to be between 5-15%, with a higher level seen among patients admitted to intensive care units, despite the evolution of antimicrobial therapy and support. (32) The CAP burden beyond the acute phase leaves several survivors with various complications, including diminished lung function, chronic respiratory

symptoms, and increased chances of cardiovascular events. Other post-CAP sequelae include recurrent pneumonia and chronic pulmonary diseases, such as bronchiectasis. (33,34)

Periodontitis as a Risk Factor of Pulmonary Infections

Increasingly, periodontitis is being established as an important risk factor for several systemic conditions, thereby revealing the complex interrelations between oral and general health. The mouth, also referred to as the gateway to the body, provides a habitat for a plethora of microbial flora whose local infection and systemic impacts can easily be felt by distant target organs and systems. In the past, oral health was seen as a local issue but increasing amounts of evidence now implicate it in systemic health. (35) Periodontitis is a chronic inflammatory disease of the supporting structure of teeth that thus provides a model for illustrating this relationship. This condition, predominantly dragged by microbial dysbiosis and aberrant immune responses, disrupts oral health and favors the occurrence of systemic diseases including cardiovascular disorders, diabetes mellitus, and respiratory infections. (19)

Systemic diseases associated with periodontitis are supported by some key mechanisms, the most important of which would be the systemic dissemination of periodontal pathogens and their products. Other examples are: *Porphyromonas gingivalis*, *Treponema denticola*, and *Tannerella forsythia*, which invade periodontal tissues, and then enter the bloodstream of the patient during normal activities such as chewing or brushing. (36) The resulting bacteremia allows these pathogens to travel to faraway organs, inflicting inflammation and tissue damage. The immune system of the host in response to these pathogens releases pro-inflammatory cytokines, such as interleukin-1 β (IL-1 β), tumor necrosis factor-alpha (TNF- α), and C-reactive protein (CRP) from the local site into the systemic circulation. These mediators then maintain a state of chronic low-grade inflammation that is an established contributor to the pathogenesis of several systemic diseases. (37)

It is recognized ever more that pulmonary infections also can be influenced by periodontitis. The aspiration of periodontal pathogens into the lower respiratory tract has been implicated in the development of pneumonia and chronic obstructive pulmonary disease (COPD). In such persons who have poor oral hygiene and advanced periodontitis, the oral cavity serves as a reservoir for respiratory pathogens. (38) On aspired admission, these pathogens could colonize the respiratory epithelium, causing infection and inflammation. *Pseudomonas aeruginosa* is a classic pathogen associated with respiratory infections and is among the pathogens identified in the mouth of patients suffering from periodontitis. (39) Dysbiosis of the oral microbiome in periodontitis allows the opportunistic pathogen to prosper, thereby increasing the risk of respiratory adverse effects.

In addition to the direct aspiration of pathogens, systemic inflammation associated with periodontitis also contributes to respiratory diseases. Pro-inflammatory mediators released from inflamed periodontal tissues can circulate to the lungs, exacerbating local inflammation and impairing pulmonary function mechanism is important for other diseases like COPD, where systemic inflammation is central to its progression. Periodontitis's chronic inflammation may also impair the respiratory immune response, resulting in increased host susceptibility to diseases.

The effects of periodontal disease on pulmonary infections are especially evident in risk groups, such as the elderly, immune-compromised individuals, and those with underlying respiratory problems. For example, hospital-acquired pneumonia-the leading cause of morbidity and mortality in hospitalized patients-has been closely associated with poor oral care and periodontitis. The interventions for oral care, such as chlorhexidine mouthwashes and professional dental cleaning, carried out in such settings decrease the incidence of pneumonia, thus reinforcing the preventive potential of these interventions in maintaining oral health.

The possible association between periodontitis and pneumonia has gained interest in recent years, owing to the knowledge that oral health embodies one component of a systemic condition. Traditionally, oral health was regarded as a localized entity; however, an increasing number of studies have documented that oral health has far-reaching consequences on general health, especially in compromised populations. The early studies suggested that poor oral hygiene was related to respiratory infections; hence, an unhealthy oral cavity may predispose individuals to infections such as community-acquired pneumonia (CAP). (40) These observations were particularly striking in institutionalized and elderly populations, where the prevalence of periodontal disease is high, and the burden of respiratory infections such as CAP is significant. Recognizing these associations thus established a rationale for further research into the putative role of periodontitis as a risk factor for pneumonia. (14)

Research on long-term care facilities and nursing homes has undoubtedly established the relationship between poor oral health and respiratory infections. The elderly in these largely compromised positions with weakened immunity and limited access to routine dental care become particularly susceptible to periodontal disease. (41) Individuals with severe periodontal disease or poor oral hygiene were found to be at a higher risk for pneumonia. Some have argued that this association has occurred partly because the oral pathogens are aspirated into the lower respiratory tract. Elderly people with swallowing difficulties or bedridden individuals are particularly susceptible to this mechanism, as oral bacteria easily are injected into a lung during micro aspiration. (42) The relevance of these findings goes beyond institutionalized

populations and promotes consideration of oral health as a modifiable risk factor for CAP in the external population.

The Mechanisms Linking Periodontitis and Pneumonia

The nature of the interrelationship between periodontitis and pneumonia is multi-pronged and gives points on how oral health can also be a factor for respiratory health. For instance, one of the most immediate paths is, through the aspiration of periodontal pathogens, into the lower respiratory tract. In periodontitis, the oral cavity provides a reservoir for pathogenic microorganisms in this instance *Porphyromonas gingivalis*, *Fusobacterium nucleatum*, and *Prevotella intermedia*. (43) After aspiration, these organisms can colonize the lungs and bring about pneumonia. The presence of periodontal pockets- a characteristic feature of periodontitis- makes this risk even greater. These pockets create an environment conducive to anaerobic bacterial growth and survival, which can then be aspirated into the respiratory tract and cause infections. (44)

The aspiration of pathogens into the lungs is direct, but the systemic inflammation that comes with periodontitis increases susceptibility to pneumonia. Periodontitis is said to be chronic inflammation due to the host's immune response toward a persistent bacterial infection. The inflammation is not limited to the oral cavity but behaves systemically. Pro-inflammatory cytokines such as interleukin-6 (IL-6), tumor necrosis factor-alpha (TNF- α), and C-reactive protein (CRP) then move freely into the systemic circulation. These inflammatory mediators potentiate pulmonary inflammation, therefore hindering the immune mechanisms responsible for the clearance of respiratory pathogens and determining lung function, hence leading to an increased risk of pneumonia. In addition, the chronic inflammation in periodontitis may affect the integrity of the respiratory epithelium and allow easier colonization by pathogens.

Periodontal disease also changes the composition of the oral microbiome, which can indirectly affect respiratory health. Normal oral health and hygiene practices maintain a delicate microbial balance in the mouths of healthy individuals. The presence of periodontitis disrupts this whole microbial balance, leading to microbial dysbiosis and pathogenic overgrowth in the oral cavity. (45) Through these changes in the oral microbiome, the respiratory tract is said to be affected whereby the aspirated oral pathogens can colonize the lung. The presence of pathogens in the lungs may result in infection and inflammation, thus paving an avenue toward pneumonia. Such a parameter is especially true for the immunocompromised category, which has compromised bacterial colonization and infection control. (46)

1.3. Contradictory Findings and Study Limitations

There is considerable indirect evidence that supports the connection of periodontal inflammation with pneumonia, as shown in various studies. (47) Nevertheless, the existing literature brings into focus many gaps and inconsistencies requiring further investigations. Findings from observational and clinical studies point toward periodontitis being a risk factor for the occurrence of community-acquired pneumonia, whereas, on the other hand, other studies have demonstrated that enhancing oral hygiene - by professional dental cleanings and antiseptic mouth rinses - is effective in minimizing the development of pneumonia in persons at high risk. (48) These results provide some indirect evidence for the contribution of periodontal health to respiratory infections and strengthen the claim that improvements in oral hygiene could protect from pneumonia.

However, contrary findings invalidate inferential conclusions regarding associations between periodontitis and pneumonia. One major limitation of these studies is the heterogeneity of study designs and their populations, and methodologies used in assessing oral health or respiratory outcomes. (49) In some studies, self-reporting measures of oral hygiene have been used, which cannot truly reflect the presence or severity of periodontitis. Other studies diagnose periodontitis using clinical examinations, and these studies may fail to control smoking and comorbidities or socio-economic interests, all of which additionally influence pneumonia risks. Not surprisingly, it is these heterogeneous designs and methodologies that stand to yield contradicting results, bringing emphasis to the call for standardization in the studied techniques targeting oral-systemic connections. (50)

Currently, literature supports a limited understanding of the causal relationship between periodontitis and pneumonia. While the association is largely established, most studies investigating causality are observational. (46) RCTs are required to confirm whether improvement in oral health translates into a reduction in respiratory infection risk by considering the effects of periodontal therapy on pneumonia occurrence. Also worthy of research are the specific mechanisms linking periodontitis to pneumonia, microbial dysbiosis and systemic inflammation that these pathways may be elucidated and potential therapeutic targets identified.

Such discoveries will not merely affect the clinical management of periodontitis but will expand into various dimensions of public health aiming at diminishing the burden of pneumonia. Introducing assessments and interventions for oral health as part of mainstream healthcare may confer considerable benefits to specific risk groups such as the aged population, the ones suffering from chronic respiratory

diseases, and those with weak immune systems. (51) Public campaigns drawing attention to oral hygiene and dental care could prove essential for pneumonia prevention and improved health outcomes.

The systematic review focuses on the complicated relationship between periodontitis and community-acquired pneumonia, thereby demonstrating the importance of oral health for respiratory health. A more profound understanding of this interaction is required to unveil possible methods of lessening the burden of the disease in the public health arena globally known as CAP. The major focus is to assess the association between periodontitis CAP-with a secondary focus being on establishing whether poor periodontal health predisposes a person to respiratory infections. Based on a systematic appraisal of the available evidence, the review aims to delineate the nature of this association and its implications for clinical practice and public health.

To reach this goal, the review also emphasizes an objective appraisal of the quality of existing research within this field. Inconsistencies in the design, method, and reporting standards of the studies have led to inconsistencies across the literature that make final conclusions difficult. An orderly evaluation of these studies will point out prospective biases and limitations such as dependent or self-reported oral health data or failure to correct several important confounding factors. This evaluation will help establish a basis for understanding the current evidence state while also becoming a springboard for future research to address the gaps identified.

An important goal will be to understand the contribution of confounding factors that may mask or exaggerate the link between periodontitis and CAP. Some variables in this regard, such as the propensity of the involved person to smoke, their age, and presence of comorbidities, have great significance with regard to both oral and respiratory health. An example of how the interplay works can be seen: Smoking is an infamous risk factor for periodontal disease, while it increases risk for respiratory infections. Aging, too, is a huge risk factor by increasing the prevalence of each condition through immune senescence factors, reduction in salivary flow, or simply lack of access to healthcare. (52) Chronic comorbidities like diabetes will further complicate this relationship since they can worsen periodontal inflammation and increase susceptibility to pneumonia. By separating these confounders, the review seeks to clarify the direct and indirect pathways through which periodontitis corresponds with CAP.

Besides confounders, the review also aims to evaluate the current body of knowledge for gaps while suggesting directions for future research. One of the obvious gaps is the lack of longitudinal studies and randomized controlled trials (RCTs) that can prove causation instead of just association. Observational studies provided valued evidence for their possible link between periodontitis and CAP; however, because

of their cross-sectional design, these studies fail to ascertain whether poor oral health precedes respiratory infections or the other way around. Moreover, many studies were done on narrow populations, such as the elderly in nursing homes, leaving it uncertain whether their findings apply to the broader population. Closing these gaps will be central for building toward an understanding of the mechanisms of this association to develop targeted interventions.

1.4. Integrating Oral Health into Public Health

The possible implications of this systematic review go beyond the mere academic-scientific interest; they point to an opportunity for acting against the burden of CAP via better oral healthcare approaches. Pneumonia has remained a major cause of morbidity and mortality across the globe, especially with vulnerable populations like the elderly, young children, and immunocompromised individuals. (53) Along these lines, periodontitis is a preventable and treatable disease; therefore, it potentially represents a modifiable risk factor that could help to decrease this burden. To demonstrate that there is a close link between periodontitis and CAP would stress the urgency of integrating oral healthcare into the respiratory health preventive agenda.

Increased awareness of systemic effects due to periodontal health may further facilitate a shift in clinical practice toward more emphasis on early detection and treatment of periodontitis. Regular dental visits and professional cleanings coupled with patient education on proper oral hygiene could help reduce the burden of periodontitis, thus lowering risks for systemic complications such as CAP. Also, antiseptic mouthwashes or chlorhexidine-based mouthwashes decrease oral cavity microbial load and may reduce the risk for aspiration pneumonia. (54) If these measures are implemented as part of public health campaigns and healthcare policies, there are opportunities to decrease CAP at the population level among high-risk groups.

The benefits of treating periodontal disease as a risk factor for CAP will reach other paradigms of public health. Reducing the prevalence of CAP by promoting oral health could save health systems a lot of money in terms of hospitalization, antibiotic use, and long-term complications of pneumonia. On the other hand, the treatment of periodontitis may have systemic advantages, given its already established associations with such diseases as cardiovascular disease, diabetes, and adverse pregnancy outcomes. This apparentness underlines the need to consider oral health not as a separate entity but an important aspect of general health and well-being.

The link between oral disease and general health carries a weighty implication for healthcare policies and practice. Oral health promotion for the reduction of respiratory infections like community-acquired pneumonia (CAP) suggests the need for interventions focused on the disorders in question. Periodontitis is preventable and treatable and is therefore a modifiable risk factor whose management could significantly reduce the burden of respiratory diseases. (55) It would be advantageous if oral health interventions, such as preventive routine dental care, patient education concerning hygiene practices, and the use of antimicrobial agents, were integrated into public health programs as preventive measures. These activities may benefit at-risk groups like the elderly, people with chronic illnesses, and those in institutionalized care; this population is disproportionately affected by both periodontitis and CAP.

There is a great potential that can be seized only if health systems adopt integrated cohesive care models for oral and systemic health. The cooperation of dental and medical professionals is vital for achieving this collaboration. In this regard, physicians, especially those working with high-risk populations, should include oral health screening and refer to potential dental issues on time. Conversely, dental professionals can identify patients at risk of possessing systemic conditions and involve other healthcare professionals in managing such risks. Advancing policies that support this interdisciplinary collaboration and imparting oral health education in conventional health care settings would be a strong stride forward. (56) For example, introducing oral health assessments during physician visits or implementing oral health interventions in the discharge protocols of hospitalized pneumonia patients could lay the groundwork for the timely detection and prevention of such diseases.

Implementing these changes, however, will require a robust database of evidence, hence the need for further research is highlighted even more urgently. While current literature points out the possible link between periodontitis and respiratory infections, there exists an enormous body of unexplored territory. Larger, better randomized controlled trials (RCTs) need to be carried out to establish causality and validate existing findings. The investigation must look into the mechanisms that mediate how periodontal health impacts respiratory outcomes, including potential microbiological and molecular pathways. Some studies could examine how the aspiration of periodontal pathogens triggers inflammation in the lungs, or they may investigate how systemic inflammation that started in the oral cavity could undermine the immune defenses.

However, research needs to expand to include other general populations so that the results can be generalized. Most of the studies conducted presently have largely been carried out on old or institutionalized individuals and hence leave considerable gaps in our understanding of the healthiest younger populations concerning their periodontal health. Another consideration is access to oral health

services differentiated by socioeconomic status since many of these underserved communities would likely be at great risk for periodontitis and systemic illnesses. Studies aimed at building an understanding of how socioeconomic factors mediate the association between oral health and respiratory infections would provide valuable baseline information for designing targeted interventions aimed at reducing health inequities.

Once some of these gaps are progressively filled, the general idea that oral health promotion translates into something much bigger for public health and clinical practice will come up much more strongly. Oral Healthcare strategies could potentially decrease the incidence of respiratory infections such as CAP leading to great savings for the health systems in forms of reduced hospitalization or antibiotic use. (57) In addition, periodontitis management is likely to be a key component of initiatives for systemic health and, therefore, may also assist in mitigating risks for chronic conditions like cardiovascular diseases and diabetic mellitus, thus amplifying their impact of improved oral health.

This research is important not only for improving health outcomes but also for bringing people together across disciplines. Dentistry must join forces with medicine, public health, and research to turn scientific findings into working strategies. Such cooperation becomes paramount for above-mentioned interventions unifying oral health with systemic health care. For example, public health campaigns can promote awareness that oral hygiene is critical for health complications beyond dental ones. In another instance, these same health providers could work together to formulate across-the-board strategies for the maintenance of oral health with respect to persons at systemic disease risk.

The need for addressing the periodontitis-CAP connection becomes urgent. Currently, respiratory infections are major causes of morbidity and mortality; therefore, reducing their burden by improving oral health would be a major target. Health-care practitioners must be cognizant of such links to prevent them from acting in a way that lessens the risk for their patients and improves their outcomes. Furthermore, policymakers must now be more aware than ever that preventing oral diseases must be an integrated approach into larger public health initiatives, including dental care access for all, regardless of socioeconomic status.

Inter-professional collaboration appears to be warranted. Dentists, physicians, and researchers should partner in investigating the intricate relationship between oral health and systemic diseases and translating this knowledge into tangible solutions. The very inclusion of periodontal disease--the risk factor for systemic diseases--presents to the health professions a great opportunity to push the agenda for prevention of diseases that enhance quality of life. An era of integrating oral health into public health policy will

transform health care delivery through prevention, early intervention, and a holistic approach to caring for patients.

So far, several studies have explored the potential link between periodontitis and community-acquired pneumonia (CAP). However, these studies were limited in scope and methodology, with many focusing on small sample sizes and fewer databases (68). As new research continues to emerge, we decided to conduct a more comprehensive systematic review to examine a larger body of evidence from a broader range of electronic databases. This systematic review aims to investigate whether periodontitis may contribute to the development of CAP in adults, considering it as a potential risk factor. Additionally, the review aims to evaluate secondary characteristics of the studies, such as study design and population demographics, to assess the consistency of findings across studies.

MATERIALS AND METHODS

1.1. Registration and protocol

A search for systematic reviews on the topic was conducted in the International Prospective Register of Systematic Reviews- PROSPERO database, and no records were found. The systematic review was registered in PROSPERO: CRD42024539085. The PRISMA statement (58) was used to guide the writing of this systematic review. The focus question was stated as follows: “In adults with periodontitis, does the presence of periodontitis increase the risk of developing community-acquired pneumonia (CAP) compared to individuals without periodontitis?”.

1.2. Selection Criteria

The systematic review had a predetermined inclusion and exclusion criteria. We would consider studies only under the following conditions: Cross-sectional, cohort, case-control and randomized studies and clinical trials in which populations of human subjects aged ≥ 18 years, received a diagnosis of periodontitis (all grades and stages), along with those exhibiting community-acquired pneumonia of any duration. (Table 1). Studies including subjects under the age of 18 or having pneumonia of any other kind were excluded.

1.3. Search strategy and search algorithm

In-depth search strategies were implemented to identify relevant studies through 18 e-databases such as PubMed, Scopus, Science Direct, Cochrane Central Register of Controlled Trials (CENTRAL), and EMBASE, as well as specific databases containing texts from gray literature, ProQuest (Appendix 1). The keywords used included “periodontitis”, “parodontosis”, “parodontoses”, “periodontosis”, “pyorrhea alveolaris”, “pneumonia”, “pneumonitis”, “lung inflammation”, “pulmonary inflammation”, “bronchopneumonia”. Searching was done afterwards, with filters applied for the last ten years for publication dates.

1.4. Data Collection and Analysis

Study Selection: Two independent reviewers (T.M. and T.A.M.) screened the titles and abstracts for relevant studies using the Rayyan tool, followed by a review of full text of eligible studies. The first research completed at 30/07/2022 (Figure 1) and an update research completed at 23/04/2024 (Figure 2). Any discrepancies or disagreements were resolved through a consensus among them. Two PRISMA flowcharts are used to depict the screening process, as shown in Figure 1 and Figure 2.

Data Extraction: Data were extracted using a standardized form that included study characteristics (design, setting, and location), population details (mean age, gender, and smoking status), variables and

outcomes (Table 1,2) A narrative synthesis was performed to summarize findings, supported by tables and figures.

1.5. Quality Assessment

The quality of the studies involved in the review were vetted by two reviewers (T.M. and T.A.M.) individually, using the ROBINS-E tool (59) for observational studies, as shown in **Figure 3 and Figure 4**. At planning stage we listed confounding factors and considered appropriateness criteria. Studies were categorized as having low, moderate or high risk of bias, on factors determined by the tool. Any disagreements were resolved by consensus. The PRISMA Checklist is mentioned in the **Annexure**.

RESULTS

A total of 5682 records were identified at the first research and 1538 records at the second one. After duplicates were removed, the titles and abstracts were examined. Of the 43 articles that were selected for complete reading, only 3 studies (60, 61, 62) met the eligibility criteria of this systematic review related to the association between periodontitis and CAP, conducted across the globe with countries including Japan, Brazil and Republic of Korea. The others were excluded for various reasons listed in **Figure 1,2**. The total sample size tallied up to 364,378 patients. **Table 3** summarizes the study characteristics. The participants were usually middle-aged, with ages ranging from 42 to 80 years old. The studies focused on development of pneumonia secondary to periodontitis. Other than the study by Awano et al. (62), the studies had an almost equal sex distribution for patients. **Table 4** summarizes the main outcome variables and findings of the studies.

Among 363,541 participants of the study Kim et al. (60), 30.1% (n =109,322) had CP. The number of CAP cases during the follow-up period was 14,838 (4.1%). The mean age of the participants was 57.8 years, and the mean age of the severe CP group was significantly higher than that of the non-CP group. The multivariable association results from Cox proportional hazard regression analyses between CP and CAP after adjustment for age, sex, household income, smoking status, alcohol consumption, physical exercise, CCI, BMI, fasting serum glucose level, and total cholesterol level, indicates that the hazard ratio (HR) for CAP was not significant in any of the models regardless of the presence or absence of CP.

Regarding the Awano et al. (62) study, the adjusted HR for mortality from pneumonia was significantly greater (HR=3.9(1.1; 13.9)) in persons with 10 or more teeth with periodontal pockets than in persons with no teeth with periodontal pockets.

As far as De Melo Neto et al. (61) study concerns, considering age, ethnicity, gender, and smoking habit, the risk of developing CAP in a patient with moderate or severe periodontitis increased to 4.4 (OR = 4.4 (1.4;13.8)).

Risk of Bias Assessment

Using the ROBINS-E tool, risk of bias assessment was carried out as shown in **Figure 3,4**. The studies lacked robust and appropriate measurement of exposure, as well as elimination of confounders in study design. There were some concerns with the studies by Kim et al. (60) and Awano et al. (62), and High risk of bias in study by De Melo Neto et al. (61).

Regarding Domain 1, the included studies are characterized as having some concerns, as they did not study the result by adjusting for all the confounding factors that may affect the outcome of the study. Also, in all 3 studies in Domain 2 the method of measuring periodontitis was not adequate. In Melo Neto et al. (61) study, the sample is very small, and the result does not seem to be representative. The overall risk of bias of this study is high.

DISCUSSION

The relationship between periodontitis and community-acquired pneumonia (CAP) has begun to receive increasing attention in recent years because of its potential implications for the oral and respiratory system. Periodontitis is an inflammatory condition that affects the supporting structures of the teeth and is said to be responsible for contributing to systemic diseases, including respiratory infections. CAP is a common, yet serious illness commonly caused by the aspiration of pathogens from the oral cavity into the lower respiratory tract and might, therefore, increase the likelihood that periodontal disease contributes to pneumonia. (69) This systematic review aims to synthesize evidence on the association between periodontitis and CAP with an emphasis on studies in Japan, Brazil, and the Republic of Korea. Though results hint at possible links, the few studies available and their methodological differences indicate that this subject should be investigated further.

4.1. Strengths and weaknesses of research

The disparity of standards for the definitions of the periodontitis and community-acquired pneumonia (CAP) from one study to the next stands in the way of a proper understanding of the relationship between the two. These heterogeneous diagnostic criteria differ from one another and thus hinder comparison and interpretation. For example, in the study by Kim et al. (60), chronic periodontitis is defined with ICD-10 codes; this gives broad clinical categorization. It is valuable in large-scale epidemiological survey studies, but it is not necessarily representative of the full spectrum or severity of the disease. ICD-10 codes make periodontitis a simple diagnosis, disregarding the clinical considerations that may affect the course or severity of the disease. In other words, the ICD-10 coding system does not differentiate, for example, between mild, moderate, or severe forms of periodontitis, thus leading to an overly simplistic understanding of the disease. While widely accessible through large datasets, this method is likely to miss important subtleties that inform a fuller understanding of the spectrum of periodontitis and its potential relationship with CAP.

In trying to classify periodontitis, De Melo Neto et al. (61) opts for a more refined classification involving clinical attachment level (CAL) and bleeding on probing (BOP), providing more insight into disease severity to categorize periodontitis, therefore, into early, moderate, and severe stages. By scoring the various levels of clinical parameters, De Melo Neto et al. (61) could bring light to the various degrees of severity, thus enhancing the possibility of defining how varying degrees of periodontitis may affect the risk of coronary artery disease. However, this classification has its shortcomings too. For instance, the clinical determinations of CAL and BOP themselves are subjective and most likely to vary according to

the expertise of the clinician, accreditation, or to the methodology adopted in a study. While somewhat more specific than ICD-10 coding, it depends on subjective measures that can be the basis for inconsistency between studies.

Awano et al. (62) consideration of the involvement of teeth and the presence of periodontal pockets in defining periodontitis is different from the other systems of classification. (63) This definition emphasizes the clinical consequences of periodontitis or periodontal disease- tooth loss, and the disease state in which pockets are formed, cavities that can harbor bacteria. The reason for this explanation lies in the idea that both the number of teeth affected and the degree of periodontal pocket depth are indicators of pathologic progression and severity. This approach therefore fits nicely with clinical practice in so far that the total number of teeth absent and severity of periodontal insult are generally useful in assessing the overall oral health of a given patient. However, it creates room for variability since various studies may use different cutoff criteria to determine what constitutes a "significant" number of teeth involved or significant depth of periodontal pockets. Limiting the definition of periodontal pockets of depth greater than 4 mm as having significance to a study population or to a specific clinical setting can introduce further discrepancies in the definitions utilized for periodontitis across studies.

Periodontitis lacks consistent definition and these varying criteria for the diagnosis of CAP employed among different studies make it more difficult to limit ambiguity when trying to form a classification system that works in all clinical settings of the disease. De Melo Neto et al. (61) are therefore making the diagnosis of CAP relying on both radiological findings and clinical criteria such as temperature elevation, increase in respiratory rate, and sputum production. This broader approach allows for a better characterization of CAP in that it blends confirmatory radiological evidence with symptoms suggestive of clinical judgment. However, this approach may be challenged by variability since not every clinical finding of CAP is universally accepted or reliably measured across studies. Irrespective, symptoms such as cough may be subjectively perceived and vary according to population studied or the standard of clinical judgment. Additionally, radiological findings may not always clearly differentiate between CAP and other types of lung infections, leading to potential misclassification.

The definitions of CAP by Kim et al. (60) could be perceived to have too little nudge for any hospitalization for even just 1 day with pneumonia. Although simple and easy to apply, it tends to leave gray areas where, perhaps, patients could have been treated and diagnosed as having CAP on an outpatient basis. The varying degrees of severity within the definition of CAP may also mean that a patient could be hospitalized with either a mild or a severe case of CAP. Thus, if the definition is based on

hospitalization alone, it fails to consider the clinical course of mild forms potentially evolving into severe forms of pneumonia.

The differences between studies for the definitions of periodontitis and of CAP limit any reliable comparisons and conclusions derived from them. Such discrepancies indicate an important gap in literature and the need for standardized diagnostic criteria in future studies. In laying down the available definitions, it is extremely difficult to identify whether any associations observed between periodontitis and CAP actually represent a causal association, or whether they simply arise from different diagnostic criteria. Different definitions just impede the possibility of data aggregation across studies or meta-analysis, both of which are vital for evidence synthesis and offer clearer perspectives on the nature of the relationship between these two clinical conditions.

The inconsistent findings observed in the current literature may additionally be explained by the variability of definitions. For example, a significant association between severe periodontitis and the onset of CAP was reported by De Melo Neto et al. (61), while Kim et al. (60) found no such association. As deduced from these observations, the definition of both periodontitis and CAP could be significantly different, which underlines the need to develop standard definitions for both diseases. This standardization could enhance the way studies can be compared and the way evidence can be derived, which in turn would help clarify the association between periodontitis and CAP.

According to De Melo Neto et al. (61), where severe periodontitis stands significant risk for CAP; those with either very severe periodontitis or moderate disease are over three times more likely to develop CAP than those with no periodontitis or very mild forms of the disease. This finding supports the proposed mechanism: that periodontal pathogens like *Porphyromonas gingivalis* or *Fusobacterium nucleatum* aspirate into the lungs from the oral cavity. This can subsequently trigger inflammatory responses in the lung, probably leading to pneumonia. This theory is supported by the understanding that oral bacteria, especially in cases of severe periodontal disease, are abundant and can easily be aspirated during swallowing or oral hygiene practices, increasing the risk of lower respiratory tract infections.

Awano et al. (62) confirmed this association by showing an elevated Hazard Ratio of 3.9 for pneumonia-related death has been observed in the case of individuals having 10 or more teeth with periodontal pockets greater than 4 mm. This emphasizes that the severity of periodontitis (depth of periodontal pockets) and extent of disease (number of affected teeth) both increase the risk of pneumonia and its fatal consequences. Deep periodontal pockets provide an ideal environment for pathogenic bacteria to flourish and infect the lower respiratory tract. The severity of periodontal disease, determined by the number of

teeth involved and the depth of pockets, appears to elevate the overall oral cavity bacterial load, which, in turn, increases the risk of aspiration and subsequent respiratory infection.

Unlike Kim et al. (60), who found no significant association between chronic periodontitis and congestive heart failure, we argue that these results reflect the experience of De Melo Neto et al. (61) and Awano et al. (62). The hazard ratios obtained in their study were quite close to 1 for all levels of severity of periodontitis; therefore, the risk of CAP was not increased. This divergence is notable and might reflect differences in, among others, study design, populations, and adjusted covariates. For example, Kim et al. (60) considered hospitalization data for CAP, so less severe episodes or those carried out in an outpatient setting may have been excluded. This would hamper the ability to highlight associations with more benign or undiagnosed types of CAP that could be affected by periodontitis. Furthermore, characteristics of the test population could have differentiated them from the other studies; this probably means that some of the risk factors of periodontitis and CAP could have been confounding. For instance, Kim et al. (60) adjusted for multiple covariates such as smoking, diabetes, and comorbidities, which could have affected the results. These negative results could also be related to the measurement problems championed by the group or to the lower amount of severe periodontitis in their sample.

The contradictory outcomes in various studies necessitate deeper explorations of these variables. Different methods, such as how periodontitis was diagnosed, how CAP was defined, and the types of adjustments for possible confounders made, can elaborate on the differences in findings. It raises a question about whether any observed association is indeed a causal association or is merely confounded by some unknown, unmeasured factors. A research approach standardizing the diagnostic criteria for both periodontitis and CAP, and incorporating longitudinal designs, would go a long way toward clarifying the strength of association and explaining in detail the mechanisms tying these two conditions.

The lack of association highlighted in Kim et al. (60) becomes even more significant when considering the large sample size of 363,541 participants in that study, a number with high statistical power that should, in principle, allow even the milder relationships between chronic periodontitis and community-acquired pneumonia (CAP) to surface. One expected a large cohort to deliver more robust conclusions, especially when compared to smaller studies with low sample sizes where the power for detection of association is often limited. In any case, Kim et al. (60) found no association that could be demonstrated between chronic periodontitis and CAP, and this remains the same despite adjusting for a diverse set of confounders, including age, smoking status, alcohol intake, comorbidities, and other recognized risk factors for pneumonia. Such extensive confounder adjustment enhances the internal validity of the study and suggests that the lack of significance is unlikely to be due to confounding alone.

The use of ICD-10 codes in diagnosing chronic periodontitis and pneumonia (also CAP), may have rendered both exposure (periodontitis) and the outcome (CAP) less sensitive and specific, even after rigorous adjustments. The ICD-10 codes were designed for administrative and billing purposes; hence, while this system provides a standardized means of classification, it fails to account for the clinical complexity that periodontitis and CAP may present. For example, chronic periodontitis ICD-10 codes do not discriminate between mild, moderate, or severe cases; they do not provide important clinical parameters, such as periodontal pocket depth or clinical attachment loss, which would have correlated with periodontal disease severity. In the same context, hospitalization-capitalized or ICD-10-coded cases of CAP may have excluded less severe pneumonia cases treated as outpatients or under different classifications, resulting in an underestimation of CAP's true burden among the population.

This imprecision with respect to definitions for exposure and outcome may account for the apparent lack of association. There is a good chance that Kim et al. (60) did not capture the full spectrum of periodontal disease with their ICD-10 codes, particularly those cases with more subtle manifestations or early stages of periodontitis that could still pose an increased risk for CAP. That said, by defining CAP based on hospitalization data, they would potentially have missed less severe cases of pneumonia managed outside the hospital, leading to an underestimation of the association between periodontitis and CAP. The absence of a significant association in Kim et al. (60) challenges the generalizability of findings from smaller studies, chiefly those that observed positive associations between periodontitis and CAP.

Another limitation of this review is that very few studies qualified for inclusion; with only three studies ultimately being included, only a minuscule evidence base was formed. The small number of studies included in this review raises major constraints to power our conclusions about the association of periodontitis and community-acquired pneumonia (CAP). The small number of studies also limits any potential generalizability across populations and settings and raises questions about potential publication bias: if results were null or inconclusive, perhaps these studies would not have been published or included. Such caution is warranted in interpreting this review's findings, and the need for large-scale, high-quality studies is apparent. Especially, studies with larger sample sizes conducted in different clinical and geographical settings would add weight to the findings and provide more secure information pertaining to a potential association between periodontitis and CAP.

One fundamental limitation is the small number of studies considered for this review, thus restricting the robustness of conclusions. Studies vary greatly, thus limit generalizability from one group to another. Inconsistencies define periodontitis and CAP across studies-another complicating factor-ICD-10 codes versus clinical measures such as clinical attachment levels, periodontal pockets, and radiological criteria.

These variations in definitions create uncertainty about whether the observed associations truly reflect a causal relationship or are merely a product of different diagnostic approaches.

This very flexibility in definitions limits comparison of results and stands yet another reason for future studies to adopt more standardized approaches. This task will include the standardization of definitions and parameters so that clear conclusions can be drawn about any cause-effect relationships between periodontitis and CAP. Large-scale studies with heterogeneous populations that incorporate methodological improvements for measuring periodontitis and pneumonia will largely clarify the relationship.

While the evidence base is limited, the studies reviewed show a commendable degree of rigor in their analyses. In particular, the studies of Awano et al. (62) and Kim et al. (60) included rigorous multiple confounders in their models, including demographic variables, smoking status, and co-morbidities, which further support the validity of their findings. This rigorous correction of potential confounders reduces the likelihood that the association was observed due to anything other than the variable studied. Smoking status, for example, is particularly important, as smoking is an established risk factor for periodontal disease as well as respiratory infections, including pneumonia. Therefore, studies, in controlling such confounders, enhance the internal validity of their results and the indication that any association observed is less likely to due to other extraneous factors. Yet, it should also be noted that although these adjusted analyses fortify the strength of the findings, they cannot offset other limitations inherent in the current evidence.

An interesting area for further research may be the correlation between the number of teeth with periodontitis and the severity of pneumonia. Awano et al. (62) found that more teeth with periodontal pockets greater than 4 mm placed an individual at higher risk for dying from pneumonia. This implies that the severity of periodontal disease—that is, the number of teeth involved—may determine the severity of respiratory infections. The pathobiology could be explained by an increased bacterial load present in the oral cavity and then aspirated into the lungs, worsening the pneumonia. We also know that periodontal disease causes systemic inflammation, which could further contribute to these adverse respiratory outcomes.

Thus, the proposed link between the severity of periodontitis as denoted by the number of involved teeth and the severity of pneumonia deserves further scrutiny. Such a correlation may also take a dose-response form whereby heavier burdens of periodontal disease translate into more severe respiratory complications. The opposite holds true. Such scenarios must be examined in other populations and contexts to ascertain

whether the alliance holds firm under varying environments and whether it can be extrapolated to other groupings of patients altogether. Additional work in this area will give valuable insight into the potential mechanisms by which periodontal disease may affect the risk and severity of pneumonia and related respiratory infections, thus serving as a basis for clinical practice to identify and treat periodontal disease early, especially in high-risk individuals for pneumonia, with the prospect of rehabilitation of patient outcome.

The limited availability of eligible studies makes systematic reviews about the association between periodontitis and community-acquired pneumonia (CAP) quite complex. The scant three studies considered in this so far narrow the opportunity of a sound meta-analysis and definite conclusion. Therefore, the demand for continued research arises to strengthen the evidence base and close these gaps. Reporting transparency in the systematic review, together with qualitative synthesis, becomes paramount in such circumstances, where data are thin, to put the study results in perspective. Nevertheless, various areas demand further in-depth inquiry in order to illuminate the periodontitis-CAP relationship.

One target will be the mechanisms involved. This scenario would entail looking at oral pathogens, such as *Porphyromonas gingivalis*, which may be aspired into the lungs and set off respiratory infections, thereby shedding light on the biological mechanisms involved. It can also be postulated that systemic inflammation and immune modulation due to periodontitis can serve to lower the threshold for pneumonia. Further study on these areas is guaranteed for a better understanding of the underlying mechanism.

"The role of confounders" is another important aspect requiring consideration. The strong possibility exists that smoking, comorbidities, and socioeconomic status contribute to observed associations, with some variability being attributed, in turn, to residual confounding. Thus, these confounders must be evaluated and investigated as confounding variables to establish overall results or, at the same time, assess whether periodontitis acts as an independent risk factor for CAP. Also, disparities in the population within the same territory, such as age, ethnicity, and availability of health care, might cause variation in findings. For instance, Kim et al. (60) focused on a predominantly Korean population, raising questions about the generalizability of their results to other regions or demographic groups.

4.2. Critical evaluation of literature

Other studies like Kelly et al. (64) and Gomes-Filho et al. (68) have investigated the association between periodontitis and respiratory diseases such as asthma, COPD, and pneumonia. Their findings

demonstrated a statistically significant association of periodontitis with the three respiratory diseases. Asthma linked most strongly, followed by pneumonia and COPD. This is also supported by a review by Almirall et al. (34) which reports poor dental health as a “definitive risk factor” for CAP. The review affirms the findings that poor periodontal health is associated with respiratory diseases and highlights that more studies are needed to clarify the diagnostic and methodological aspects. On the other hand, an umbrella review undertaken by the Canadian Dental Hygienist’s Association (65) found that variations in study populations, CHX concentrations, frequency of administration, and other oral care interventions served to compromise the evidence available to them. Some studies had a positive effect on outcomes such as ICU mortality and length of stay, but these were mainly observed in patients following cardiac surgery. The Bradford Hill criteria for causality were applied, and in the end, their study could not establish the causal contribution of periodontal microbes to respiratory diseases such as pneumonia.

Similarly, a review by Cagnani et al. (66) showed the included studies show a positive link between periodontal disease and aspiration pneumonia, but the conclusions were limited due to inconsistencies in methodologies, small sample sizes, suboptimal control for confounding factors, and varying diagnostic criteria. Therefore, in view of the limitations that exist, the current evidence cannot adequately support the claim that periodontal disease constitutes an absolute risk factor according to this study. However, this is not enough to completely disregard the association as Yang et al. (67) found that Cox regression analysis revealed that pneumonia incidence was significantly lower in the periodontal treatment group over an average follow-up of about seven years.

4.3. Future research

Establishing a clear temporal association between periodontitis and CAP presents a challenge in observational studies. Current evidence only tentatively supports causal inference because the cross-sectional design would not allow confirmation as to whether periodontitis in fact temporally precedes the development of CAP in the study subjects. Longitudinal cohort studies are of paramount necessity to determine the true directionality of the relationship, thus enabling a far better understanding of potential causal links between the two disorders. Observational studies alone will not provide strong at first sight evidence for causal inference. An approach that could demonstrate whether periodontal treatment is beneficial in preventing CAP would also prove whether oral health control could be used as a possible preventive strategy against respiratory infections. Such studies would have increased importance for public health if it is determined that periodontitis is a risk factor for CAP. Inclusion of oral health within broader preventive strategies for respiratory diseases would be a cost-effective measure to reduce the burden exerted by pneumonia, especially among high-risk populations.

Right now, it is a priority that future studies find a way to overcome the limitations faced in the present ones. Harmonizing definitions of periodontitis and CAP, increasing sample sizes, and diversifying study populations would benefit the generalizability of the results. It is also worth investigating whether specific subgroups would be targeted, such as the old or the immunocompromised, more prone to both periodontitis and respiratory infection. All these challenges, tackled in future studies, could help create a stronger base of evidence and provide much clearer answers in regard to the affiliation between periodontitis and CAP.

CONCLUSION

Three studies having a total population of 364,378 were reviewed for the study about the association between periodontitis and community-acquired pneumonia (CAP) in Japan, Brazil, and South Korea. The results were varied: Awano et al. (62) and De Melo Neto et al. (61) found a higher risk of pneumonia-related mortality and CAP in individuals with severe periodontitis, while Kim et al. (60) found no significant association. The study calls for further research to minimize methodological and clinical heterogeneity in larger, longitudinal-based studies and intervention trials for improved understanding of the potential role of oral health in respiratory infections.

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FIGURES AND TABLES

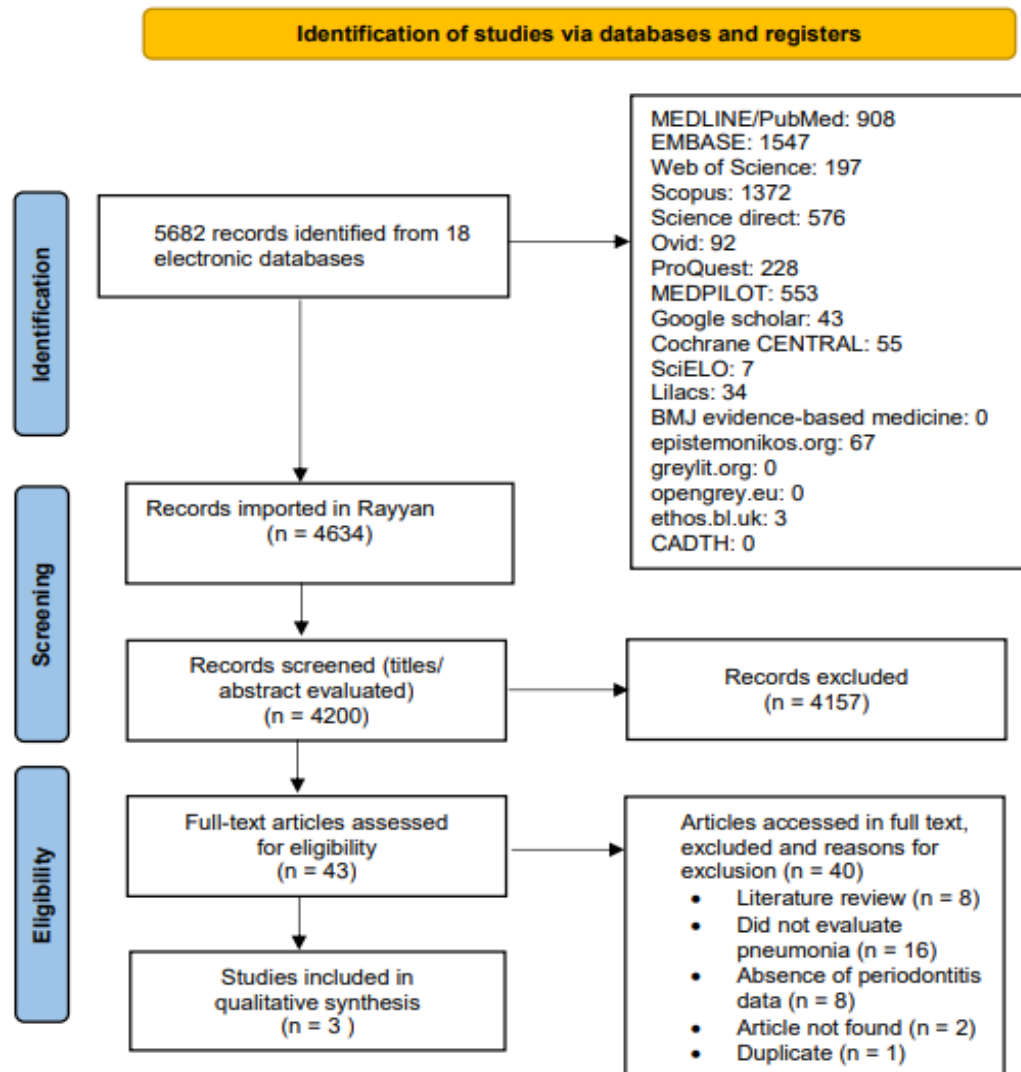


Figure 1. PRISMA Flowchart 1st research (30/07/2022)

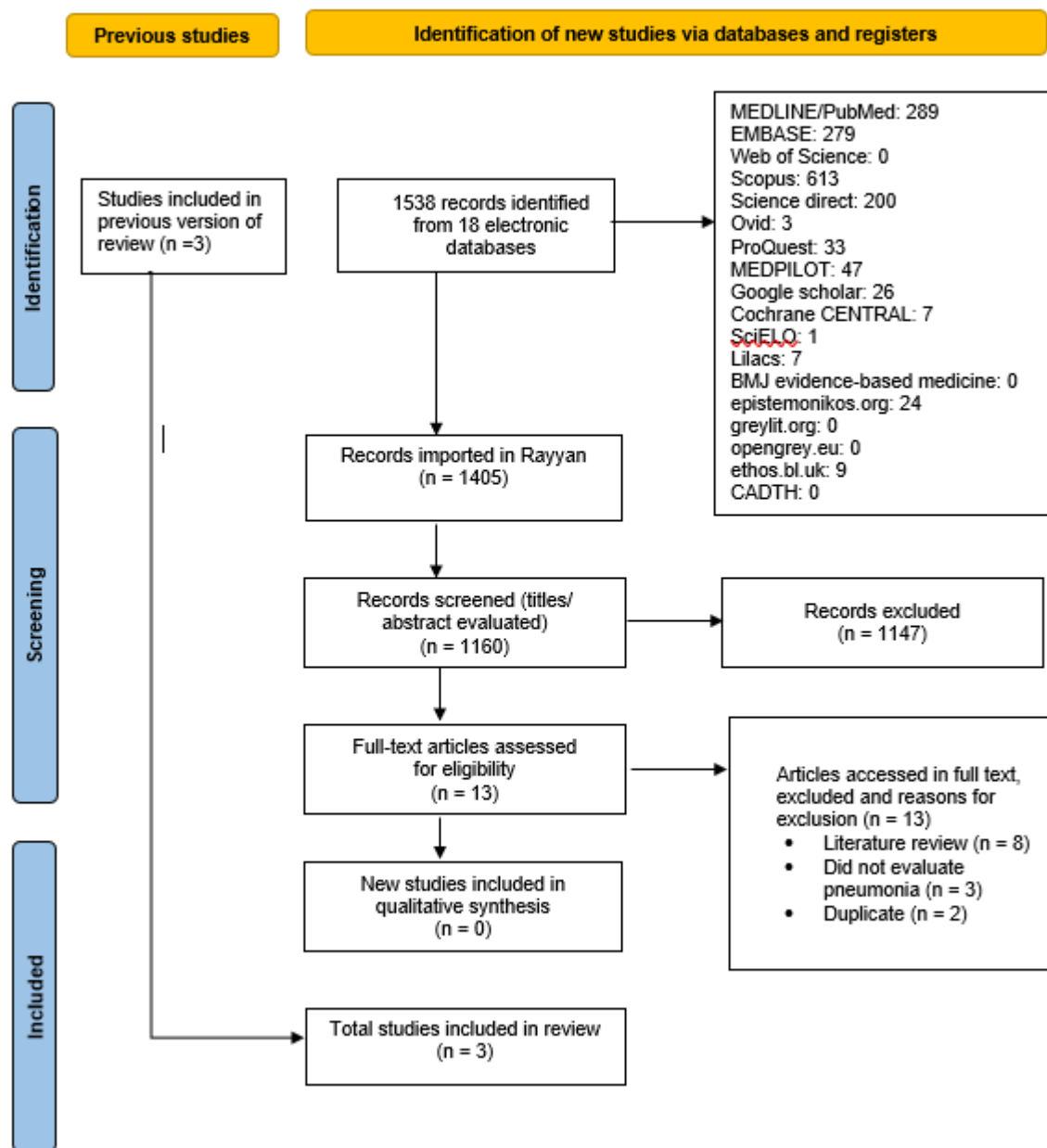


Figure 2. PRISMA flowchart update research 23/04/2024

Figure 3. ROBINS-E summary plot.

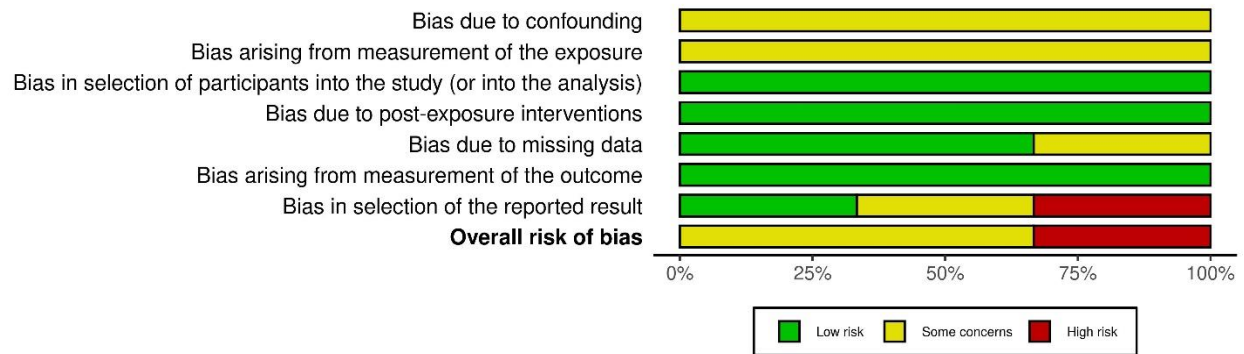


Figure 4. ROBINS-E traffic light plot.

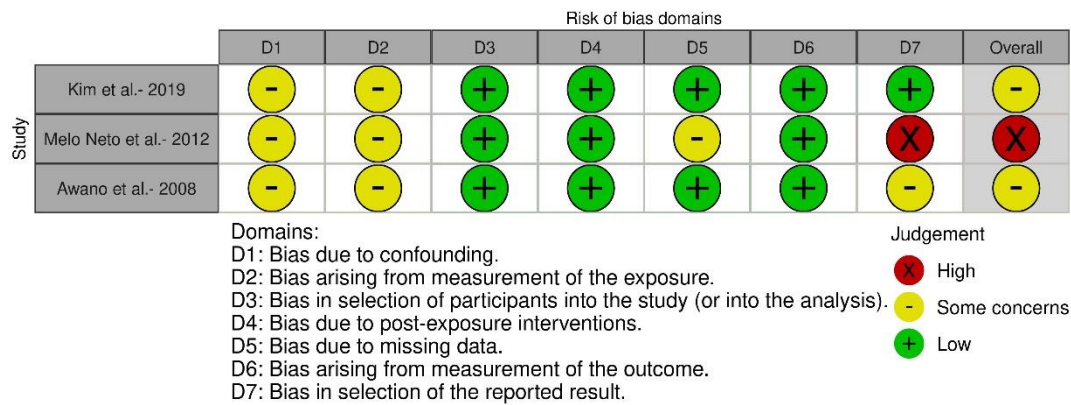


Table 1. PECOS components of the study's research question

Population	Patients aged ≥ 18 years with a diagnosis of periodontitis
Exposure	Diagnosis of periodontitis of all grades and stages
Comparison	Subjects without periodontitis (periodontal health)
Outcome	Community- acquired pneumonia
Studies	Cross-sectional, Cohort, Case-control studies or RCTs

Table 2. Study Characteristics

Study	Country	Study Design	Sample Size	Age Range of Subjects	Population
Awano et al. (2008)	Japan	Cohort	697	80	Cases: patients with pneumonia-associated death Control: -Gr.1: Alive patients -Gr.2: patients with other causes associated
De Melo Neto et al. (2013)	Brazil	Case-control	140 (cases=70, controls=70)	Cases: 42.0 ± 15.6 Controls: 44.1 ± 17.3	Cases: patients with community-acquired pneumonia Controls: patients without community-acquired pneumonia
Kim et al. (2019)	Republic of Korea	Cohort	363,541	Non-CP= 57.6 ± 8.8 CP = 57.56 ± 8.16	Cases with Community Acquired Pneumonia with and without Chronic Periodontitis

Table 3. Main outcomes and findings

Study	Periodontitis Variable	CAP variable	Summary of main findings	Results on association	Adjusted covariates
Awano et al. (2008)	Oral health questionnaire Number of teeth	Systemic health questionnaire	The mean number of teeth with periodontal pockets was significantly greater in the pneumonia-death group than in other groups	Four-year mortality from pneumonia and number of teeth with pockets >4mm (vs. no pockets): 1-9 teeth with PD>4mm: HR=1.1(0.4; 3.0) 10≤ teeth with PD >4 mm: HR=3.9(1.1; 13.9)	Gender, history of stroke, serum total cholesterol, BMI
De Melo Neto et al. (2013)	Chronic periodontitis (CP): at least one of six sites with CAL ≥1mm and BOP present Severity of CP: -Early: CAL<3 mm -Moderate: CAL≥3mm and <5 mm -Severe: CAL ≥5 mm PD, CAL, BOP	Radiological evidence of pneumonia in a chest radiograph and at least one of the following clinical findings (temperature ≥37.8° C, respiratory rate >25 cycles per minute, presence of sputum, heart rate >100 beats per minute, crackles, decreased breath sounds, myalgia and night sweats)	Early periodontitis was not significantly associated with the risk for developing CAP Severe periodontitis was significantly associated with a higher risk of suffering CAP, when compared with periodontally healthy subjects	Early periodontitis and community-acquired pneumonia: OR = 1.8 (0.3;11.0) Moderate/severe periodontitis and community-acquired pneumonia: OR = 4.4 (1.4;13.8)	Age, ethnicity, gender and smoking habit
Kim et al. (2019)	Chronic Periodontitis as defined by the ICD-10 codes (K05.30, K05.31, K05.32, K05.38, K05.39)	Hospitalization for ≥1 day due to CAP	CP is not a potential risk factor for CAP. The etiology and causality between periodontal disease and CAP remain unclear, and little is known about the underlying mechanism	Hazards Ratio for CAP via CP: - Mild CP = 0.95(0.89-1.01) - Moderate CP = 1.03(0.94-1.12) - Severe CP = 1.00(0.96-1.04)	Age, sex, household income, smoking status, alcohol consumption, physical exercise, CCI, BMI, fasting serum glucose and total cholesterol level

ANNEXURE

Table 4. PRISMA Checklist 2020

Section and Topic	Item #	Checklist item	Location where item is reported (page)
TITLE			
Title	1	Identify the report as a systematic review.	2
ABSTRACT			
Abstract	2	See the PRISMA 2020 for Abstracts checklist.	7
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of existing knowledge.	15
Objectives	4	Provide an explicit statement of the objective(s) or question(s) the review addresses.	15
METHODS			
Eligibility criteria	5	Specify the inclusion and exclusion criteria for the review and how studies were grouped for the syntheses.	20
Information sources	6	Specify all databases, registers, websites, organisations, reference lists and other sources searched or consulted to identify studies. Specify the date when each source was last searched or consulted.	20
Search strategy	7	Present the full search strategies for all databases, registers and websites, including any filters and limits used.	20
Selection process	8	Specify the methods used to decide whether a study met the inclusion criteria of the review, including how many reviewers screened each record and each report retrieved, whether they worked independently, and if applicable, details of automation tools used in the process.	20
Data collection process	9	Specify the methods used to collect data from reports, including how many reviewers collected data from each report, whether they worked independently, any processes for obtaining or confirming data from study investigators, and if applicable, details of automation tools used in the process.	20
Data items	10a	List and define all outcomes for which data were sought. Specify whether all results that were compatible with each outcome domain in each study were sought (e.g. for all measures, time points, analyses), and if not, the methods used to decide which results to collect.	20
	10b	List and define all other variables for which data were sought (e.g. participant and intervention characteristics, funding sources). Describe any assumptions made about any missing or unclear	20

Section and Topic	Item #	Checklist item	Location where item is reported (page)
		information.	
Study risk of bias assessment	11	Specify the methods used to assess risk of bias in the included studies, including details of the tool(s) used, how many reviewers assessed each study and whether they worked independently, and if applicable, details of automation tools used in the process.	21
Effect measures	12	Specify for each outcome the effect measure(s) (e.g. risk ratio, mean difference) used in the synthesis or presentation of results.	Not Applicable
Synthesis methods	13a	Describe the processes used to decide which studies were eligible for each synthesis (e.g. tabulating the study intervention characteristics and comparing against the planned groups for each synthesis (item #5)).	20
	13b	Describe any methods required to prepare the data for presentation or synthesis, such as handling of missing summary statistics, or data conversions.	Not Applicable
	13c	Describe any methods used to tabulate or visually display results of individual studies and syntheses.	Not Applicable
	13d	Describe any methods used to synthesize results and provide a rationale for the choice(s). If meta-analysis was performed, describe the model(s), method(s) to identify the presence and extent of statistical heterogeneity, and software package(s) used.	20
	13e	Describe any methods used to explore possible causes of heterogeneity among study results (e.g. subgroup analysis, meta-regression).	Not Applicable
	13f	Describe any sensitivity analyses conducted to assess robustness of the synthesized results.	Not Applicable
Reporting bias assessment	14	Describe any methods used to assess risk of bias due to missing results in a synthesis (arising from reporting biases).	21
Certainty assessment	15	Describe any methods used to assess certainty (or confidence) in the body of evidence for an outcome.	Not Applicable
RESULTS			
Study selection	16a	Describe the results of the search and selection process, from the number of records identified in the search to the number of studies included in the review, ideally using a flow diagram.	22
	16b	Cite studies that might appear to meet the inclusion criteria, but which were excluded, and explain why they were excluded.	38
Study characteristics	17	Cite each included study and present its characteristics.	22
Risk of bias in studies	18	Present assessments of risk of bias for each included study.	22

Section and Topic	Item #	Checklist item	Location where item is reported (page)
Results of individual studies	19	For all outcomes, present, for each study: (a) summary statistics for each group (where appropriate) and (b) an effect estimate and its precision (e.g. confidence/credible interval), ideally using structured tables or plots.	41
Results of syntheses	20a	For each synthesis, briefly summarise the characteristics and risk of bias among contributing studies.	40
	20b	Present results of all statistical syntheses conducted. If meta-analysis was done, present for each the summary estimate and its precision (e.g. confidence/credible interval) and measures of statistical heterogeneity. If comparing groups, describe the direction of the effect.	Not Applicable
	20c	Present results of all investigations of possible causes of heterogeneity among study results.	Not Applicable
	20d	Present results of all sensitivity analyses conducted to assess the robustness of the synthesized results.	Not Applicable
Reporting biases	21	Present assessments of risk of bias due to missing results (arising from reporting biases) for each synthesis assessed.	22
Certainty of evidence	22	Present assessments of certainty (or confidence) in the body of evidence for each outcome assessed.	Not Applicable
DISCUSSION			
Discussion	23a	Provide a general interpretation of the results in the context of other evidence.	24
	23b	Discuss any limitations of the evidence included in the review.	24
	23c	Discuss any limitations of the review processes used.	Not Applicable
	23d	Discuss implications of the results for practice, policy, and future research.	31
OTHER INFORMATION			
Registration and protocol	24a	Provide registration information for the review, including register name and registration number, or state that the review was not registered.	20
	24b	Indicate where the review protocol can be accessed, or state that a protocol was not prepared.	20
	24c	Describe and explain any amendments to information provided at registration or in the protocol.	Not Applicable
Support	25	Describe sources of financial or non-financial support for the review, and the role of the funders or sponsors in the review.	Not Applicable
Competing interests	26	Declare any competing interests of review authors.	Not Applicable

Section and Topic	Item #	Checklist item	Location where item is reported (page)
Availability of data, code and other materials	27	Report which of the following are publicly available and where they can be found: template data collection forms; data extracted from included studies; data used for all analyses; analytic code; any other materials used in the review.	37